YONGGANG JIANG

GitHub: yonggangjiang   Google Scholar: Yonggang Jiang
Max Planck Institute for Informatics (MPI-INF), E14 Room 315, Saarland Informatics Campus
E-mail: yjiang@mpi-inf.mpg.de   Homepage: yonggangjiang.github.io

EDUCATION

Max Planck Institute for Informatics and Saarland University *Apr. 2023 – Present*
PhD in Computer Science
Advisor: Danupon Nanongkai

Max Planck Institute for Informatics and Saarland University *Oct. 2021 – Mar. 2023*
Preparatory Phase for Doctoral Study

Nanjing University *Sep. 2017 – Jul. 2021*
B.S. in Computer Science and Technology
GPA: 94.4/100, Ranking: 1/31

University of California, Berkeley *Jan. 2020 – May 2020*
Berkeley International Study Program
GPA: 4.0/4.0

HONORS & AWARDS

Google PhD Fellowship *2025*
Distinguished Paper Award, SPAA 2025 *2025*
Outstanding Graduate, Nanjing University *2021*
National Elite Program Scholarship, Outstanding Prize (top 1) **2018, 2019, 2020**
China Collegiate Programming Contest (CCPC), Gold Medal *2018*
National Olympiad in Informatics in Provinces (NOIP), First Prize *2015*

PUBLICATIONS

***Co-authors are listed in alphabetical order.**

[1] **A Subquadratic Two-Party Communication Protocol for Minimum Cost Flow**
Hossein Gholizadeh, Yonggang Jiang.
In submission.

[2] **Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth**
Yonggang Jiang, Changki Yun.
In submission.

[3] **Perfect Simulation of Las Vegas Algorithms via Local Computation**
Xinyu Fu, Yonggang Jiang, Yitong Yin.
In submission.

[4] **Reviving Thorup's Shortcut Conjecture**
Aaron Bernstein, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.
In submission.

[5] **Reducing Shortcut and Hopset Constructions to Shallow Graphs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

[6] Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs
Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

[7] Minimum s-t Cuts with Fewer Cut Queries
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[8] Shortcuts and Transitive-Closure Spanners Approximation
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[9] New Oracles and Labeling Schemes for Vertex Cut Queries
Yonggang Jiang, Merav Parter, Asaf Petruschka.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[10] Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.

[11] Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances
Jan van den Brand, Hossein Gholizadeh, Yonggang Jiang, Tijn de Vos.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2025.
Distinguished Paper Award

[12] Deterministic Vertex Connectivity via Common-Neighborhood Clustering and Pseudorandomness
Chaitanya Nalam, Yonggang Jiang, Thatchaphol Saranurak, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

[13] Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
Joakim Blikstad, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

[14] Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights
Vikrant Ashvinkumar, Aaron Bernstein, Nairen Cao, Christoph Grunau, Bernhard Haeupler, Yonggang Jiang, Danupon Nanongkai, Hsin Hao Su.
European Symposium on Algorithms (ESA) 2024.

[15] Finding a Small Vertex Cut on Distributed Networks
Yonggang Jiang, Sagnik Mukhopadhyay.
ACM Symposium on Theory of Computing (STOC) 2023.

[16] Robust and Optimal Contention Resolution without Collision Detection
Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2022.

[17] Tight Trade-off in Contention Resolution without Collision Detection
Haimin Chen, Yonggang Jiang, Chaodong Zheng.

ACM Symposium on Principles of Distributed Computing (PODC) 2021.

RESEARCH VISITS & INVITED TALKS

National University of Singapore, Singapore, ***Sep. 2025***
Hosted by Yi-Jun Chang.
Talk: Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work

New York University, New York, NY, USA ***Sep. 2025***
Hosted by Aaron Bernstein.
Talk: Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work

Carnegie Mellon University, Pittsburgh, PA, USA ***Aug. 2025***
Hosted by Jason Li.
Talk: Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work

University of Michigan, Ann Arbor, MI, USA ***July. 2025***
Hosted by Thatchaphol Saranurak.

ETH Zurich, Zurich, Switzerland ***Feb. 2025***
Hosted by Sorrachai Yingchareonthawornchai.

Tsinghua Universtiy, Beijing, China ***Dec. 2024***
Hosted by Ran Duan.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

Nanjing University, Nanjing, China ***Dec. 2024***
Hosted by Yitong Yin.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

Shanghai University of Finance and Economics, Shanghai, China ***Dec. 2024***
Hosted by Pinyan Lu.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

ETH Zurich, Zurich, Switzerland ***Nov. 2024***
Hosted by Weiming Feng.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

INSAIT, Sofia, Bulgaria ***Sep. 2024***
Hosted by Bernhard Haeupler.

University of Warwick, Coventry, UK ***Jun. 2024***
Hosted by Sayan Bhattacharya.
Talk: Deterministic k-Vertex Connectivity in k Max-Flows

University of Sheffield, Sheffield, UK ***Jun. 2024***
Hosted by Sagnik Mukhopadhyay.

Nanjing University, Nanjing, China ***Nov. 2023***
Hosted by Penghui Yao.

Rutgers University, New Brunswick, NJ, USA ***Jun. 2023***
Hosted by Aaron Bernstein.

Aalto University, Helsinki, Finland ***Apr. 2023***
Hosted by Parinya Chalermsook.

Aalto University, Helsinki, Finland *Dec. 2022*
Hosted by Parinya Chalermsook.

University of Sheffield, Sheffield, UK *May 2022*
Hosted by Sagnik Mukhopadhyay.

CONFERENCE & WORKSHOP TALKS

Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances
SPAA 2025, Portland, OR, USA *July. 2025*

Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
STOC 2025, Prague, Czech Republic *Jun. 2025*

Contributed Talk: Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
HALG 2025, Zurich, Switzerland *Jun. 2025*

Contributed Talk: Deterministic k-Vertex Connectivity in k Max-Flows
HALG 2024, Warsaw, Poland *Jun. 2024*

Parallel, Distributed, and Quantum Exact Single-Source Shortest Paths with Negative Edge Weights
ESA 2024, London, UK *Jun. 2024*

Finding a Small Vertex Cut on Distributed Networks
STOC 2023, Orlando, FL, USA *Jun. 2023*

Robust and Optimal Contention Resolution without Collision Detection
SPAA 2022, online *Jul. 2022*

Tight Trade-off in Contention Resolution without Collision Detection
PODC 2021, online *Jul. 2021*

TEACHING EXPERIENCE

Seminar: Foundations of Machine Learning, MPI-INF *Summer 2023*
Role: Coordinator and Speaker.

Data Structures and Algorithms, Nanjing University *Fall 2020, Fall 2019*
Role: Teaching Assistant.

STUDENTS & MENTORING

Ioannis Dorkofikis (M.Sc.) *Spring 2025*
Internship at MPI-INF, from NTUA, Greece.

Changki Yun (B.Sc.) *Fall 2024*
Internship at MPI-INF, from SNU, Korea.
Outcome: Paper on parallel vertex connectivity (in submission)

Hossein Gholizadeh (B.Sc.) *Summer 2024*
Bachelor's Thesis Internship at MPI-INF, from KIT, Germany.
Outcome: Paper in SPAA 2025 (Distinguished Paper Award)

Alireza Danaei (B.Sc.) *Summer 2023*
Internship at MPI-INF, from Sharif Univ., Iran